

MASTER'S THESIS DEFENSE

Demand Forecasting and Inventory Optimisation Using Machine Learning

*A simulation-based evaluation on the M5-Forecasting (Walmart) dataset, with an
SME-accessible Power BI dashboard*

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What This Defense Covers

01



The Business Problem

Why forecasting alone doesn't solve inventory decisions

02



Methodology

Five model tiers, OUTL policy, and forward simulation

03



Results

Forecast accuracy and the central inventory finding

04



Recommendations

A diagnostic tool for SME decision-makers

Every Retailer Faces the Same Trade-off

Too little stock

→ lost sales, eroded customer trust

Too much stock

→ tied-up cash, holding cost, spoilage risk

Large retailers solve this with dedicated data-science teams and enterprise software. Small and medium-sized enterprises (SMEs) typically have neither — yet face an identical operational problem.

Published claims say:

10–25%

forecast accuracy improvement

5–15%

inventory cost reduction — from published ML forecasting studies

Four Structural Gaps in Current Practice



GAP 1

Forecast $\neq$ Decision

Most studies report accuracy and stop — they never translate forecasts into inventory outcomes.



GAP 2

No Live Validation

Inventory performance is usually assumed analytically, not measured by running the policy forward.



GAP 3

SMEs Left Behind

Enterprise tools assume infrastructure and staffing SMEs simply don't have.



GAP 4

Uncertainty Ignored

Textbook safety-stock formulas assume symmetric errors — real retail demand is skewed and intermittent.

Five Objectives Guided This Research

- 1 Comparatively evaluate forecasting approaches — classical, ML, deep learning, ensemble, and probabilistic — on a hierarchical retail benchmark.
- 2 Develop a systematic feature-engineering strategy capturing the principal drivers of retail demand.
- 3 Integrate the best-performing forecast with the Order-Up-To-Level (OUTL) inventory policy.
- 4 Test whether literature-reported gains (10–25% accuracy, 5–15% cost) transfer to SKU-daily granularity.
- 5 Deliver a Power BI dashboard translating outputs into an accessible tool for SME decision-makers.

A Consistent Finding, With One Critical Caveat

ML and ensemble methods consistently outperform classical statistics
across retail forecasting studies — this is well replicated.



The caveat that matters most:

The published 10-25% / 5-15% benchmarks are measured on data aggregated to the weekly or category level — not the daily, individual-product level where real inventory decisions are made.

This thesis was designed to directly test whether those benchmark figures survive the move to SKU-daily granularity, using real product-store-day data and inventory outcomes measured — not assumed.

M5-Forecasting (Walmart) Dataset

3,049

products

10

stores, 3 US states

1,913

days (2011–2016)

502

series stratified
subsample used

56

day out-of-sample
test window

Chosen for scale, hierarchical structure mirroring real SME retail operations, and status as an established academic benchmark. A stratified subsample (proportional across category $\times$ state) was used for computational tractability.

Five Model Tiers, Compared Head-to-Head



Tier 1

Classical

MA, SES, AutoARIMA,
Croston, Croston-SBA,
Seasonal Naive



Tier 2

Standard ML

Random Forest,
XGBoost, LightGBM
(Tweedie loss)



Tier 3

Deep Learning

LSTM — 128/64
stacked units,
dropout 0.2



Tier 4

Ensemble

Stacking: LGBM + XGB
+ RF → Ridge
meta-learner



Tier 5

Probabilistic

LightGBM quantile
regression
(P10 – P99)

Trained strictly on historical data. Evaluated exclusively on a later, entirely unseen test period.

Translating Forecasts Into Inventory Decisions

The Order-Up-To-Level (R, s, S) Policy

A periodic review policy: every R days, if the inventory position falls at or below reorder point s, an order is placed to raise stock to order-up-to level S.

Three policy variants compared — differing only in how safety stock is calculated:

Classical

— Gaussian formula fed by classical forecast

ML-Gaussian

— Gaussian formula fed by ML ensemble forecast

ML-Empirical-Quantile

— distribution-aware formula from quantile model

Safety Stock Formulas

$$SS = z \cdot \sigma \cdot \sqrt{R + L}$$

Gaussian (textbook) — assumes symmetric forecast-error distribution

$$SS = (P95 - P50) \cdot \sqrt{R + L}$$

Empirical quantile — reflects the true, skewed shape of retail demand

Measuring Performance, Not Assuming It



Most Published Studies

Calculate expected cost and service level analytically — using a formula to project what should happen under an assumed stockout rate.



This Thesis

Runs each policy forward, day by day, against real 56-day test demand for all 502 products — genuinely tracking stock, genuinely recording every stockout.

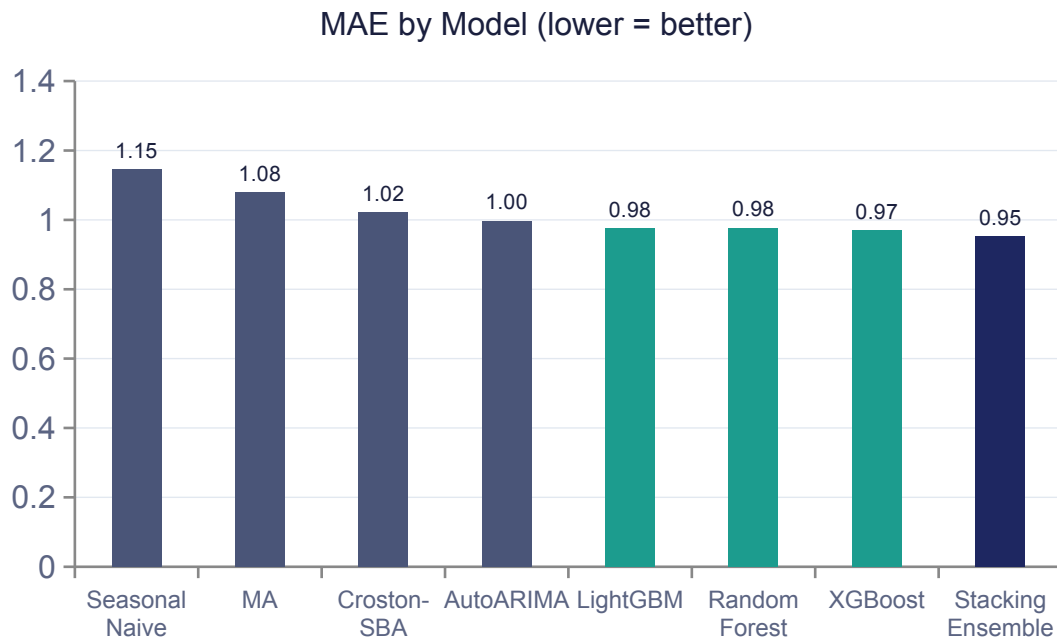
Falsification Test: Can It Detect a Stockout?

Before trusting the simulation, it was deliberately stress-tested under conditions engineered to force stockouts.

Stress Condition	Realised SL	Stockout Cost
Real buffers (baseline)	100.0%	\$0
Safety stock set to zero	74.0%	\$124,338
Demand scaled ×3	94.1%	\$87,691
Replenishment suppressed	0.0%	\$469,556

Every stress condition correctly registered degraded service and rising cost — confirming the 100% baseline result reflects demand genuinely being met, not a counter that never fires.

The Ensemble Wins, But the Margin Is Modest



+4.5%

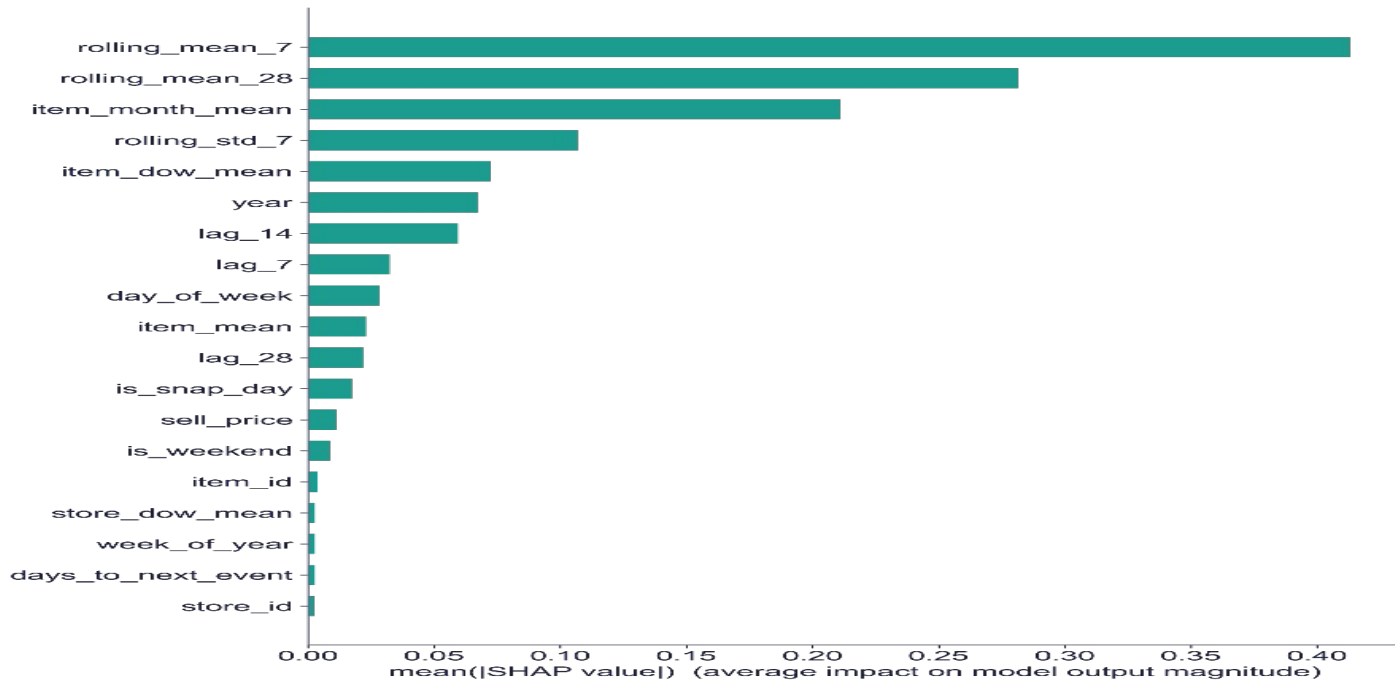
MAE improvement vs best classical baseline

+9.5%

RMSE improvement — large-error / spike sensitivity

MAPE improvement was essentially flat — below the 10–25% literature benchmark, as anticipated (Slide 6).

What Actually Drives the Forecast



Top 5 features explain 78.6% of model behaviour — dominated by recent sales trend and item-level seasonality, not opaque signals.

THE CENTRAL FINDING

100%

Realised Service Level

— achieved by ALL THREE policies (Classical, ML-Gaussian, ML-Quantile) across every one of the 502 products, over the full 56-day test window. No policy ever stocked out.

Consequence:

With zero stockouts, there was no stockout cost left for a smarter forecast or a smarter safety-stock formula to save.

ML-Quantile policy cost:

+1.5%

more than the classical baseline —
unnecessary safety stock, no offsetting benefit

Why the Earlier 21% Estimate Did Not Survive

Earlier (Analytical) Draft

21%

Assumed the Gaussian policy achieved ~83% service and the quantile policy ~95% "by calibration," then computed cost from that gap.

The quantile policy won by construction — before any simulation ran.

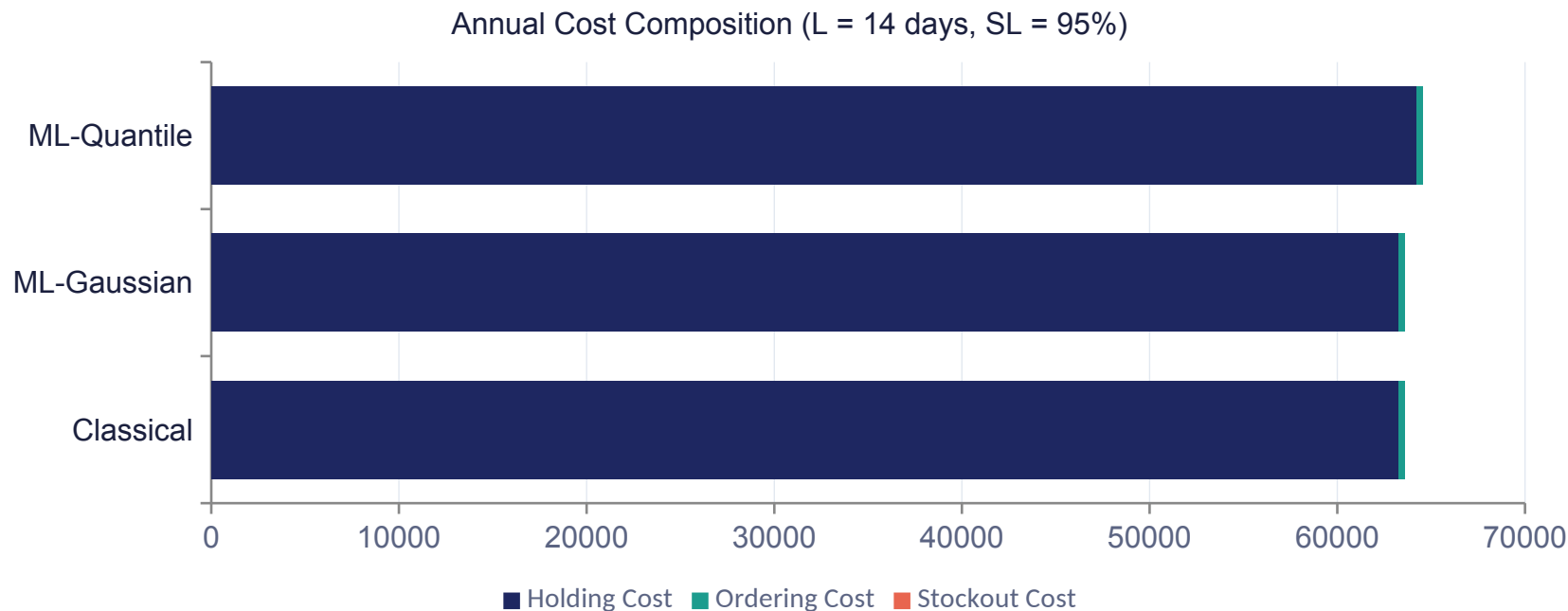
This Thesis (Simulated)

–1.5%

Both policies actually achieve 100% service — the quantile buffer protects against stockouts that never occur.

The Ch1 5–15% inventory-cost target is not achieved on this dataset — treated as a finding, not a failure.

Holding Cost Dominates; Stockout Cost Is Zero



Bars start at \$0 — the ML-Quantile difference is a genuine but small ~1.5% increase, not the large gap a truncated axis would suggest.

The Finding Is Robust — Not a One-Off Result



Across Lead Time (7–21 days)

–1.3% to –1.7%

Quantile policy consistently more expensive, never favourable



Across Stockout Multiplier (0.4×–2.0×)

No change

Multiplier never engages — zero stockouts to price at any level

A threefold demand-shock stress test (Slide 11) shows even that is not enough, on this dataset, to make the quantile policy clearly superior.

A Boundary Condition, Not a Blanket Claim

The value of ML forecasting and distributional safety stock is conditional on the demand regime:



Genuine stockout risk exists

→ the larger, empirically calibrated buffer earns back its holding cost by preventing real stockouts.



Service is already saturated

→ the buffer protects against stockouts that don't occur
— pure added cost, no benefit. (The M5 test regime.)

An Accessible Power BI Dashboard for SMEs

0.952

MAE Ensemble

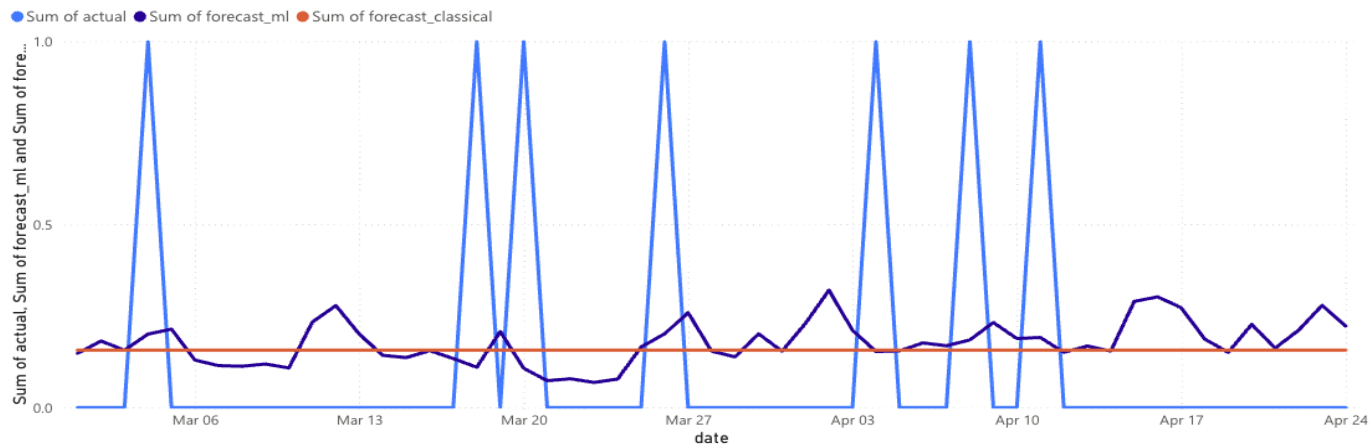
100.0%

SL Quantile

-1.5%

Cost Chg. Quantile

Sum of actual, Sum of forecast_ml and Sum of forecast_classical by date



Cost Analysis — The Finding, Made Visible

Cost Analysis - Holding, Ordering, and Stockout Components

Total Annual Cost

64.57K

Sum of total_cost

Holding Cost

64.24K

Sum of holding_cost

Ordering Cost

331.82

Sum of ordering_cost

Stockout Cost

0

Sum of stockout_cost

lead_time

14

policy

ML_EMP_QUANTILE

KPI tiles: ML-Empirical-Quantile policy, L = 14, m = 1.0 (central case, matches Ch5 Table 5.2)

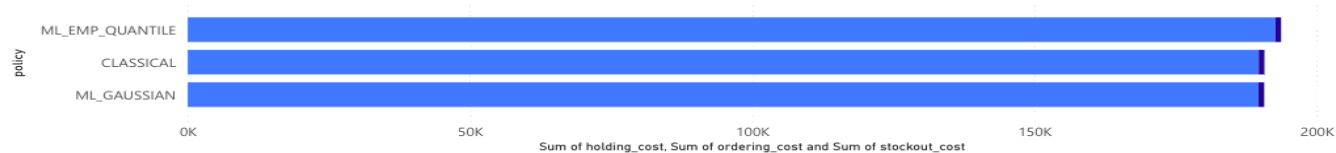
policy	Average of realised_sl_pct	Sum of total_cost
CLASSICAL	100.00	758,748.24
ML_EMP_QUANTILE	100.00	770,189.76
ML_GAUSSIAN	100.00	758,605.71
Total	100.00	2,287,543.71

Total Annual Cost(\$) by Lead Time x Policy

lead_time	CLASSICAL	ML_EMP_QUANTILE	ML_GAUSSIAN	Total
7	183,060.51	185,473.89	183,030.45	551,564.85
10	186,374.55	189,033.93	186,341.43	561,749.91
14	190,762.50	193,718.25	190,725.66	575,206.41
21	198,550.68	201,963.69	198,508.17	599,022.54
Total	758,748.24	770,189.76	758,605.71	2,287,543.71

Cost Composition by Policy (L=14)

● Sum of holding_cost ● Sum of ordering_cost ● Sum of stockout_cost



Free (Power BI free tier). Runs on a standard laptop. Requires only data a typical POS system already produces.

Theoretical and Practical Contributions

Theoretical

- First simulation-based measurement of the Gaussian-vs-empirical safety-stock choice on M5
- Demonstrates ML forecasting's inventory value is regime-conditional, not automatic
- Shows the inventory comparison is robust to the stockout-cost multiplier (zero stockouts at $0.4\times$ – $2.0\times$)

Practical

- End-to-end, reproducible, free-and-open-source pipeline (< 3 hrs on a laptop)
- A forward-simulation engine usable as a pre-investment diagnostic
- An interactive Power BI dashboard for non-technical SME managers

For SMEs: Diagnose Before You Invest

1

Don't trust published % claims alone

Ask: at what data granularity, and was cost measured or assumed?

2

Run the diagnostic on your own data first

The delivered simulation engine tells you if you face real stockout risk

3

Invest where the diagnostic says risk is real

High-margin, fast-moving, or highly seasonal categories are the best candidates

4

Deploy the dashboard regardless

Free, low-risk visibility — independent of which forecasting method feeds it

What This Study Does Not Claim

Limitations

- Single 56-day test window — a longer or differently-timed window could expose real stockouts
- LSTM evaluated at one architecture only — more elaborate sequence models untested
- Stockout cost modelled as linear — real-world costs (e.g. churn) may be non-linear

Future Work

- Re-test on retail benchmarks with more volatile demand (Rossmann, Favorita)
- Validate the boundary condition in a live SME deployment with genuine stockout exposure
- Extend the simulation engine to non-linear stockout costs and lead-time uncertainty

The Value of ML Forecasting Is Real — But Conditional, Not Automatic

This thesis delivers both the evidence for that conclusion and a practical, free, open-source tool that lets any organisation test which side of that condition it falls on — using its own data, before spending anything further.

The barrier this research removes is not financial. It is informational.

Thank You

Questions & Discussion

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